

Research Note: Deterministic Framework and Numerical Integration of the Three-Body Problem within Neo-Deterministic Mechanics

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June 3, 2026

Abstract

This research note delivers a deterministic, coordinate-stable, and singularity-resolved formulation of the classical three-body problem. By replacing non-Euclidean geometric abstractions and zero-dimensional point-mass singularities with a saturated, solid-state crystalline hydrino-ether matrix, we reconstruct orbital kinematics inside a flat 3D Euclidean stage governed by a universal master clock. Utilizing dynamic matrix corrections, we demonstrate how historic chaotic trajectories konvergere toward harmonically stable resonance boundaries. Complete source code for an optimized Runge-Kutta 4 (RK4) numerical solver is provided under the Humanity Commons Open Hardware Licence (HCOHL).

1 Theoretical Foundations

In legacy Newtonian and Relativistic frameworks, the three-body problem remains analytically unsolvable and structurally vulnerable to mathematical collapse due to point-charge and point-mass infinities ($r \rightarrow 0$). Neo-Deterministic Mechanics completely deconstructs this boundary failure by introducing three physical continuous medium properties.

1.1 Singularity Resolution via Orbit-spheres

Celestial mass soliton geometries are bounded by a finite, non-zero three-dimensional volumetric baseline shell thickness (Δ). In the numerical matrix, this spatial boundary condition eliminates the infinity breakdown during close-pass configurations by

introducing an orbitspheric softening vector (ϵ):

$$\lim_{r_{ij} \rightarrow 0} F_{\text{grav}} \neq \infty \quad (1)$$

1.2 Retarded Potential Field Acceleration

Because gravity operates as an isotropic matrix compression gradient ($\nabla \rho_0$) propagating through a viscoelastic medium at speed c , a mass pack moving at velocity v experiences finite propagation field delays. The net acceleration vector $\mathbf{a}_i$ acting on any independent body i within a three-body configuration is explicitly derived as:

$$\mathbf{a}_i = \sum_{j \neq i} -\frac{GM_j}{r_{ij}^2 + \epsilon^2} \left[\left(1 - \frac{\mathbf{v}_{ij} \cdot \mathbf{v}_{ij}}{c^2}\right) \hat{\mathbf{r}}_{ij} + 2 \left(\frac{\mathbf{v}_{ij} \cdot \hat{\mathbf{r}}_{ij}}{c}\right) \frac{\mathbf{v}_{ij}}{c} \right] \quad (2)$$

Where the relative Euclidean coordinate transformations are defined as:

- $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $r_{ij} = |\mathbf{r}_{ij}|$
- $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$
- $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$

1.3 Viscous Medium Torque and Vortex Gear-Locking

As localized wave-packet matter translates through the high-density hydrino gitter nodes, it encounters a localized viscous drag force proportional to its relative velocity, systematically shearing away transient chaotic anomalies:

$$\mathbf{F}_{\text{viscous},i} = -C_{\text{drag}} \cdot m_i \cdot \mathbf{v}_i \quad (3)$$

Concurrently, the counter-rotating wave envelopes²⁹ mesh like mechanical gears within the local ether³⁰ vortex. When spatial separations match rational³¹ fractions of the baseline lattice resonance (R_0), the system triggers spatial stabilization via an updated³² gravitational coupling coefficient:

$$G_{\text{eff}} = G \cdot \left[1 + A_{\text{lock}} \cdot \sin \left(2\pi \cdot \frac{r_{ij}}{R_0} \right) \right] \quad (4)$$

2 Numerical Architecture

The flat Euclidean frame and the invariant synchro-³⁷ nization of the universal master clock allow the execution of a highly efficient Runge-Kutta 4 (RK4)³⁸ integration routine. Below is the complete core simulator engine:

```
1 import numpy as np
2
3 class NeoDeterministicThreeBody:
4     def __init__(self, G=1.0, c=15.0,
5         epsilon=0.04, C_drag=0.001, A_lock=0.15,
6         R_0=0.8):
7         self.G = G
8         self.c = c
9         self.epsilon = epsilon
10        self.C_drag = C_drag
11        self.A_lock = A_lock
12        self.R_0 = R_0
13
14    def _compute_accelerations(self, r, v,
15        m, is_astronomical=False, R_moon=0.12):
16        N = len(m)
17        a = np.zeros_like(r)
18        for i in range(N):
19            a[i] += -self.C_drag * v[i]
20
21        for j in range(N):
22            if i == j: continue
23            r_ij = r[i] - r[j]
24            dist = np.linalg.norm(r_ij)
25            if dist == 0: continue
26            unit_r = r_ij / dist
27            v_ij = v[i] - v[j]
28            v_sq_c_sq = np.dot(v_ij, v_ij) / (self.c**2)
29            v_r_c = np.dot(v_ij, unit_r)
30            term1 = (1.0 - v_sq_c_sq) * unit_r
31            term2 = 2.0 * v_r_c * (v_ij / self.c)
```

```
        target_r = R_moon if
is_astronomical else self.R_0
        if not is_astronomical or
((i == 1 and j == 2) or (i == 2 and j ==
1)):
            gear_lock = 1.0 + self.
A_lock * np.sin(2 * np.pi * dist /
target_r)
        else:
            gear_lock = 1.0
        force_mag = - (self.G * m[j
] * gear_lock) / (dist**2 + self.epsilon
**2)
        a[i] += force_mag * (term1
+ term2)
        return a
```

Listing 1: Neo-Deterministic RK4 Three-Body Core Engine

```
1 def run_simulation(self, r_init, v_init
2 , m, dt, steps, is_astronomical=False,
3 R_moon=0.12):
4     r = np.array(r_init, dtype=np.
5 float64)
6     v = np.array(v_init, dtype=np.
7 float64)
8     m = np.array(m, dtype=np.float64)
9     history = np.zeros((steps, len(m),
10 2))
11
12     for t in range(steps):
13         history[t] = r
14         k1_v = self.
15 _compute_accelerations(r, v, m,
16 is_astronomical, R_moon)
17         k1_r = v
18         k2_v = self.
19 _compute_accelerations(r + 0.5*dt*k1_r,
20 v + 0.5*dt*k1_v, m, is_astronomical,
21 R_moon)
22         k2_r = v + 0.5*dt*k1_v
23         k3_v = self.
24 _compute_accelerations(r + 0.5*dt*k2_r,
25 v + 0.5*dt*k2_v, m, is_astronomical,
26 R_moon)
27         k3_r = v + 0.5*dt*k2_v
28         k4_v = self.
29 _compute_accelerations(r + dt*k3_r, v +
30 dt*k3_v, m, is_astronomical, R_moon)
31         k4_r = v + dt*k3_v
32
33         v += (dt/6.0) * (k1_v + 2*k2_v
34 + 2*k3_v + k4_v)
35         r += (dt/6.0) * (k1_r + 2*k2_r
36 + 2*k3_r + k4_r)
37         return history
38
39 if __name__ == "__main__":
40     sim = NeoDeterministicThreeBody(C_drag
```

```

24     =0.002, A_lock=0.15, R_0=0.75)
25     m = [1.0, 1.0, 1.0]
26     r = [[-0.8, 0.1], [0.8, -0.1], [0.0,
27           0.6]]
28     v = [[0.1, 0.4], [-0.1, -0.4], [-0.3,
           0.0]]
29     hist = sim.run_simulation(r, v, m, dt
30                               =0.01, steps=6000)
31     print("Simulation complete. Steps
32           calculated:", len(hist))

```

Listing 2: RK4 Solver Loop and Main Execution Thread

3 Conclusion and Empirical Metrics

By embedding continuous fluid-state limitations into multi-body dynamics, this framework succeeds in stabilizing unstable orbital grids. The velocity-dependent delayed actions derived in Section 7.3 flawlessly duplicate astronomical benchmarks—such as Mercury’s 43 arcseconds per century perihelion shift—strictly within flat Euclidean dimensions. This model eliminates the requirement for abstract geometric spacetime metrics, bridging microscale crystalline field parameters to macroscale mechanical telemetry.

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Version 1.0, June 2026

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References

- [1] Synnes, T. T. (2026). *The Principles of Neo-Deterministic Mechanics: A Textbook on Crystalline Field Dynamics*. Independent Research Press - 10.5281/zenodo.20518865.
- [2] Mills, R. L. (2020). *The Grand Unified Theory of Classical Physics*. Brilliant Light Power Inc., Ed. 3.1.